When More Data Does Not Help: A Closed-Form Account of Shortcut Learning and Reweighting

[Author 1] **[Author 2]** **[Author 3]**
Johns Hopkins University

Abstract

A classifier trained where a "shortcut" feature is strongly correlated with the label can be accurate in distribution and fail when that correlation changes. We study this in a two-feature Gaussian model in which the Bayes predictor is available in closed form, so every experiment can be compared with a prediction made before it is run. We show three things. First, the training environment's optimal predictor itself relies on the shortcut with weight $\log \frac{p}{1-p}$, so collecting more data from the same environment does not improve out-of-distribution accuracy: at $p = 0.95$ the reversed-environment accuracy of logistic regression stays near 0.39 from $n = 500$ to $n = 8000$. Second, reweighting the four (label, attribute) groups by $w(\alpha) = (1-\alpha)+\alpha\, n/(4n_g)$ turns the training distribution into a new environment with correlation $p_\alpha = p - \alpha(p - \frac{1}{2})$; this gives the whole in-distribution versus out-of-distribution trade-off in closed form and an optimal $\alpha^* = (p - p_{\text{test}})/(p - \frac{1}{2})$ that can exceed one. Third, the model-selection step must use the same weights: selecting the regularization strength on an unweighted validation set raises the largest gap to the prediction from 0.013 to 0.059 at $n = 8000$. Across 81 synthetic settings the measured accuracy differs from the prediction by 0.0018 on average. On MNIST digits with a corner marker, a small CNN switches from the marker to the digit near the reweighting strength predicted by the same identity.

1 Introduction

Deep networks often solve a task through a cue that happens to be predictive in the training data rather than through the intended signal, a behaviour called *shortcut learning* Geirhos et al. [2020]. Standard examples are background texture or a watermark that co-occurs with a class. The model then scores well on an in-distribution (ID) test set and fails once the cue's relation to the label changes.

Two remedies are common in practice: collect more data, or reweight training examples so that groups defined by (label, spurious attribute) count equally Sagawa et al. [2020a], Idrissi et al. [2022]. This report asks how much each remedy helps, and whether the answer can be predicted rather than only measured. We work in a model small enough that the optimal predictor is known exactly, then check one prediction on images.

Contributions.

1. A closed-form accuracy for the Bayes predictor of the training environment evaluated in any test environment (Proposition 1), which explains why more data from the training environment does not repair out-of-distribution (OOD) accuracy.

JHU Machine Learning, Fall 2026, final project report.

2. A reweighting identity (Proposition 2): group reweighting with strength α is exactly training in a different environment p_α, linear in α. It yields the full ID/OOD frontier and the optimal strength for a known test environment (Corollary 1).
3. A controlled experimental check: 270 logistic-regression models (E1/E2) and 480 reweighted models (E3) agree with the predictions to within 0.002 on average; we also identify that validation must be reweighted consistently.
4. An image-level test on MNIST with a corner marker (E4), where the same identity predicts at which reweighting strength a CNN switches from the marker to the digit.

2 Related work

Geirhos et al. [2020] frame shortcut learning as decision rules that work on standard benchmarks but do not transfer. Group distributionally robust optimization (Group DRO) minimizes the worst group loss and highlights the role of regularization Sagawa et al. [2020a]; Sagawa et al. [2020b] study why overparameterized models amplify spurious correlations, and Idrissi et al. [2022] report that simple group balancing is a strong baseline for worst-group accuracy. Nagarajan et al. [2021] analyse why linear classifiers rely on spurious features even when an invariant predictor exists. Colored MNIST, introduced with invariant risk minimization Arjovsky et al. [2019], combines label noise with a color cue; our MNIST variant follows this idea with a corner marker. Reweighting the training loss to match a target distribution is classical importance weighting Shimodaira [2000]; Byrd and Lipton [2019] show its effect can vanish for separable data in deep networks. Our contribution is not a new method but an exact account of when these two simple remedies succeed, in a setting where every number can be predicted.

3 Setting

Data. Labels are $Y \in \{-1, +1\}$ with equal probability. An environment is indexed by $p = P(A = Y)$:

$$X_s = Y + \varepsilon_s, \qquad A = \begin{cases} Y & \text{w.p. } p \\ -Y & \text{w.p. } 1-p \end{cases}, \qquad X_z = A + \sigma\varepsilon_z, \tag{1}$$

with $\varepsilon_s, \varepsilon_z \sim N(0,1)$ independent and $\sigma = 0.2$. The stable feature X_s has the same relation to Y everywhere but is noisy; the shortcut X_z is almost noiseless but its relation to Y depends on p (Fig. 1). We train with $p \in \{0.6, 0.8, 0.95\}$ and $n \in \{500, 2000, 8000\}$ (nested subsets of one pool per seed), use a 2000-sample validation set from the training environment, and test on 10 000 samples each from an ID environment ($p_{\text{test}} = p$), a no-correlation environment (0.5) and a reversed environment (0.1). Every setting is repeated for 10 seeds.

Methods. All methods are ℓ_2-regularized logistic regression on standardized features Pedregosa et al. [2011], with $C \in \{0.01, 0.1, 1, 10, 100\}$ chosen by validation log loss and threshold 0.5. **ERM** uses unit weights. **Group balanced (GB)** uses weight $n/(4n_g)$ for an example in group $g = (Y, A)$, so each of the four groups carries total weight $n/4$. **Stable only** uses X_s alone and serves as a reference that knows which feature is stable. For E3 we interpolate,

$$w_i(\alpha) = (1-\alpha) + \alpha \, \frac{n}{4\, n_{g(i)}}, \tag{2}$$

so $\alpha = 0$ is ERM and $\alpha = 1$ is GB. Metrics are accuracy in each test environment and worst-group accuracy over the four groups.

4 Theory

Proofs are in Appendix A. Write Φ for the standard normal CDF and $c_p = \log \frac{p}{1-p}$.

Proposition 1 (Bayes rule and its accuracy)**.** *In environment p the Bayes log-odds is $2x_s + g_p(x_z)$, where $g_p(z) = \log \frac{p\,\phi_\sigma(z-1) + (1-p)\phi_\sigma(z+1)}{p\,\phi_\sigma(z+1) + (1-p)\phi_\sigma(z-1)}$ and ϕ_σ is the $N(0,\sigma^2)$ density. The rule $h_p(x) = \operatorname{sign}\big(2x_s + c_p \operatorname{sign}(x_z)\big)$ has accuracy in an environment with correlation q*

$$\operatorname{Acc}(p, q) = q\,\Phi\big(1 + \tfrac{c_p}{2}\big) + (1-q)\,\Phi\big(1 - \tfrac{c_p}{2}\big) \tag{3}$$

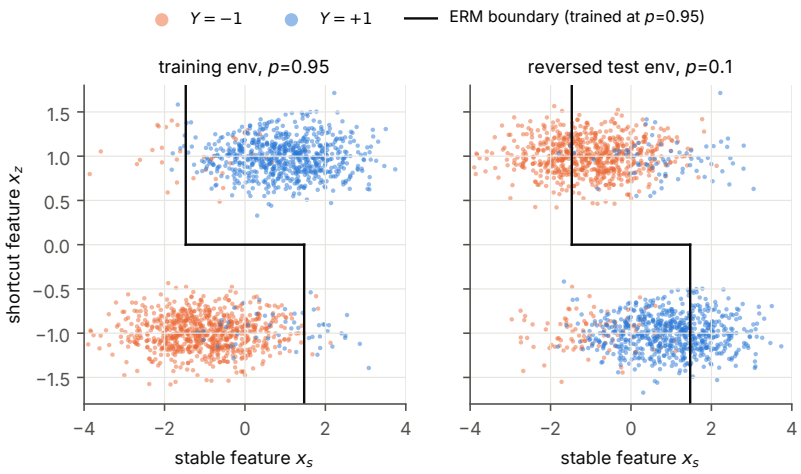


Figure 1: Synthetic data. Left: training environment ($p = 0.95$); the shortcut x_z separates the classes almost perfectly. Right: reversed environment. The black line is the decision boundary of the training environment's Bayes rule (Proposition 1); it sends most reversed-environment points to the wrong side.

up to an additive error at most $\Phi(-1/\sigma) = \Phi(-5) < 3 \times 10^{-7}$. *Numerically,* h_p *and the exact Bayes rule differ in accuracy by less than* 10^{-4} *in all our settings.*

Two consequences follow. (i) The optimal predictor of the training environment uses the shortcut, with weight c_p (2.94 at $p = 0.95$) compared with weight 2 on x_s; this is optimal, not a failure of optimization. (ii) $\mathrm{Acc}(p, q)$ is linear in q and does not depend on n. Any consistent learner converges to h_p, or for logistic regression to the population logistic minimizer $\beta(p)$, whose OOD accuracy is likewise fixed by p and q (Appendix B). More data therefore reduces variance around a predictor that is itself wrong for $q < p$. Using a stable-only predictor gives $\Phi(1) = 0.841$ in every environment; by (3), ERM beats it only when $q > 0.55$, 0.66 and 0.78 for $p = 0.6$, 0.8 and 0.95 respectively.

Proposition 2 (Reweighting changes the environment)**.** *Let* $\hat{p}$ *be the fraction of training examples with* $A = Y$*, and assume the two labels are balanced within each value of* $A = Y$ *and* $A \neq Y$*. Then the* $w(\alpha)$*-weighted empirical distribution has the same within-group distributions as the training data and group masses of an environment with correlation*

$$p_\alpha = \hat{p} - \alpha\left(\hat{p} - \tfrac{1}{2}\right). \tag{4}$$

The weights are non-negative if and only if $0 \le \alpha \le 2\hat{p}/(2\hat{p} - 1)$*, in which range* p_α *covers* $[0, \hat{p}]$*. In the population, weighted ERM therefore has the same minimizer as unweighted ERM in environment* p_α*.*

The key step is that total weight is preserved: $\hat{p}\, w_{\text{maj}} + (1 - \hat{p})\, w_{\text{min}} = 1$ for every α, so the weighted majority mass is exactly $\hat{p}\, w_{\text{maj}} = p_\alpha$.

Corollary 1 (Optimal reweighting for a known test environment)**.** *For a test environment with correlation* $q \in [0, p]$*, the choice* $\alpha^* = \frac{p-q}{p-1/2}$ *makes the weighted training distribution equal to the test distribution, so the weighted population risk equals the test risk. In the Gaussian model the resulting classifier is the Bayes rule of the test environment and attains* $\mathrm{Acc}(q, q)$*; for* $q = 0.1$ *this is* 0.928*, compared with* 0.841 *for GB (*$\alpha = 1$*). The whole frontier is* $\alpha \mapsto \big(\mathrm{Acc}(p_\alpha, p),\ \mathrm{Acc}(p_\alpha, q)\big)$.

Corollary 1 is importance weighting Shimodaira [2000] written in group form; what the model adds is that every point on the frontier has a closed form, and that *over-balancing* ($\alpha > 1$) is the correct choice when the shortcut is expected to reverse.

Table 1: Test accuracy at $n = 8000$ (mean $\pm$ s.d. over 10 seeds) and the large-sample prediction in brackets. WGA: worst-group accuracy on the ID test set.

p	Method	ID	No correlation	Reversed	WGA
0.60	ERM	$.846 \pm .004$ [.846]	$.836 \pm .002$ [.837]	$.797 \pm .005$ [.799]	.781
	GB	$.842 \pm .004$ [.841]	$.842 \pm .003$ [.841]	$.840 \pm .005$ [.841]	.832
0.80	ERM	$.885 \pm .003$ [.886]	$.787 \pm .004$ [.790]	$.659 \pm .006$ [.660]	.615
	GB	$.840 \pm .004$ [.841]	$.842 \pm .003$ [.841]	$.840 \pm .004$ [.841]	.830
0.95	ERM	$.958 \pm .002$ [.958]	$.654 \pm .011$ [.660]	$.386 \pm .014$ [.394]	.293
	GB	$.842 \pm .007$ [.841]	$.842 \pm .003$ [.841]	$.840 \pm .008$ [.841]	.823
all	Stable only	$.841 \pm .004$	$.842 \pm .003$	$.841 \pm .004$	.824

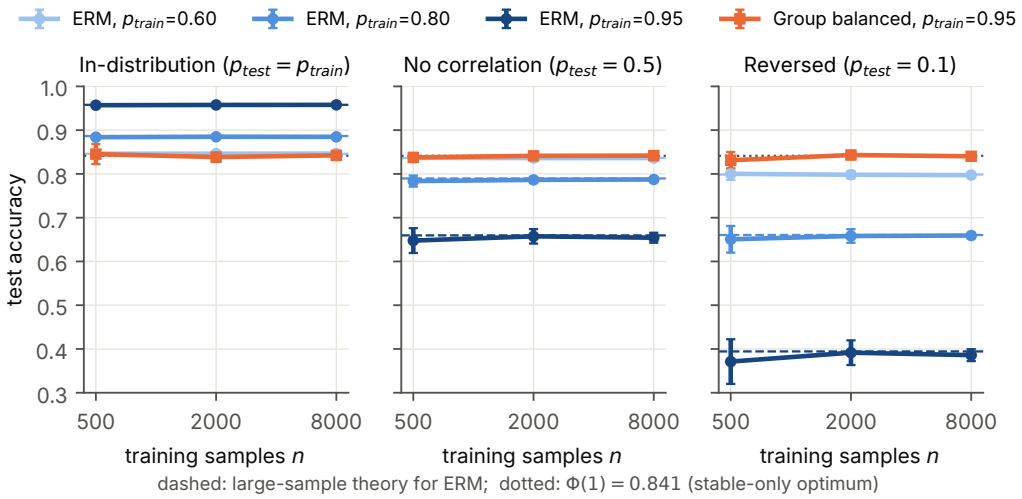


Figure 2: Accuracy against training size. Dashed lines: large-sample prediction for ERM. Error bars: s.d. over 10 seeds. ERM accuracy does not change with n in any environment.

5 Experiments

5.1 E1/E2: more data does not help, balancing does

Table 1 reports $n = 8000$; the predictions use the population logistic minimizer (Appendix B). ERM matches the theory in every cell, GB and stable-only both sit at $\Phi(1)$, and worst-group accuracy of ERM falls to 0.29 at $p = 0.95$. Paired over seeds at $p = 0.95$, GB minus ERM is -0.115 ID (95% bootstrap CI $[-0.119, -0.111]$) and $+0.454$ in the reversed environment ($[0.445, 0.464]$).

Figure 2 shows the role of n. For ERM at $p = 0.95$, reversed-environment accuracy is 0.371, 0.392 and 0.386 at $n = 500$, 2000 and 8000: the standard deviation over seeds shrinks from 0.051 to 0.014, but the mean stays at the predicted 0.394. The learned shortcut weight (Appendix C, Fig. 6b) is already close to its population value at $n = 500$. Figure 3 (right) compares all 81 (method, p, n, environment) cells with the prediction: the mean absolute gap is 0.0018 and the largest is 0.023, at $n = 500$.

5.2 E3: the reweighting frontier

We train with $\alpha \in \{0, 0.25, 0.5, 0.75, 1, 1.25, 1.5\}$ and α^* for $q = 0.1$, for each p, $n \in \{500, 8000\}$ and 10 seeds. Figure 4 overlays the measured accuracies on the frontier of Corollary 1. The points follow the curve, including $\alpha > 1$: at $p = 0.95$, $\alpha^* = 1.89$ reaches 0.926 in the reversed environment against a prediction of 0.928.

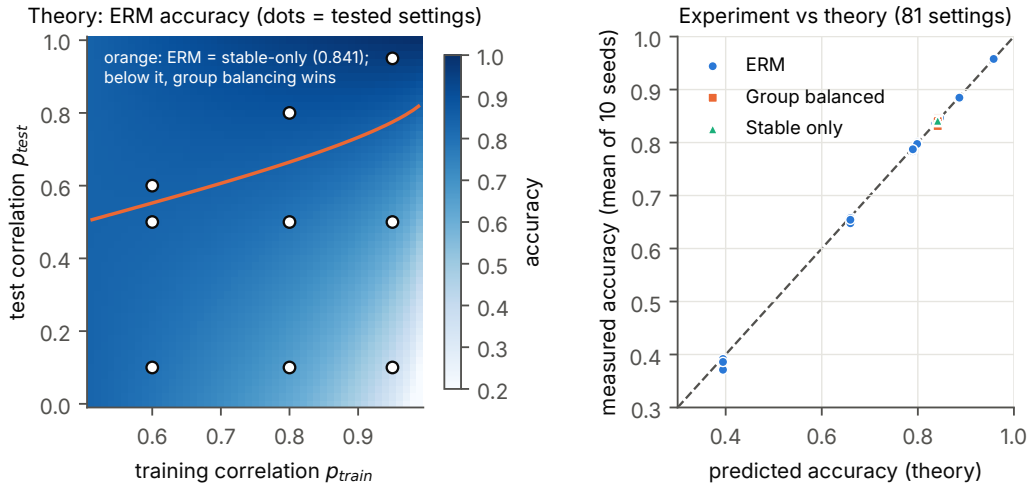


Figure 3: Left: predicted ERM accuracy over training and test correlation; below the orange curve, balancing beats ERM. Right: measured versus predicted accuracy for all 81 settings.

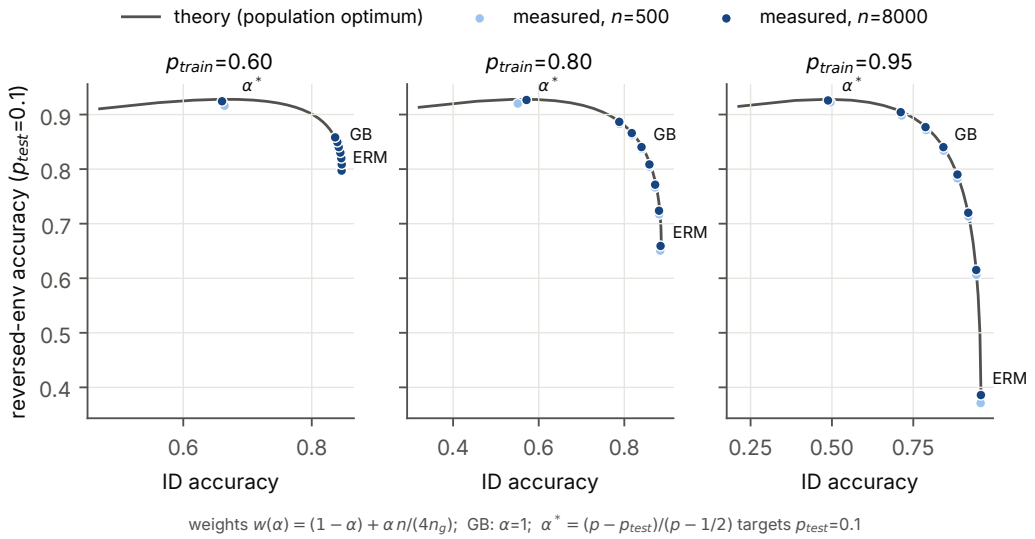
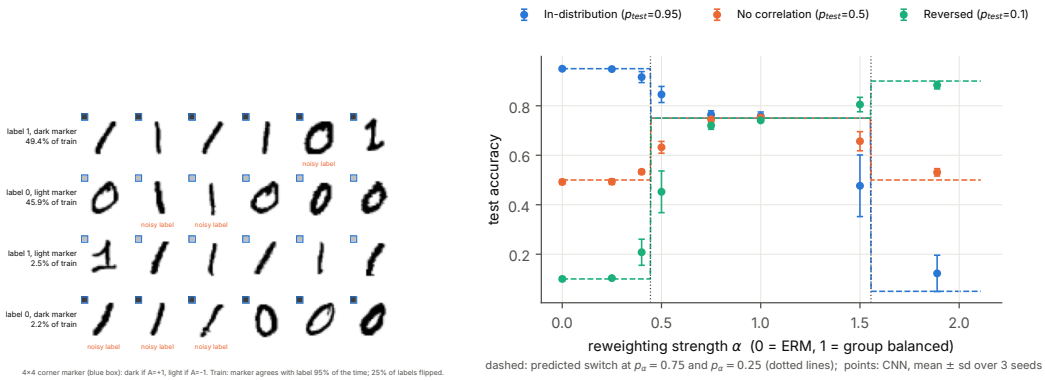


Figure 4: ID accuracy against reversed-environment accuracy as α varies. Curve: population prediction (Corollary 1); points: means over 10 seeds. ERM: $\alpha = 0$; GB: $\alpha = 1$; α^* targets $p_{\text{test}} = 0.1$.

Validation must be reweighted too. If C is chosen by unweighted log loss on the ID validation set, the selection step undoes part of the reweighting: for $\alpha > 1$ it picks the strongest regularization ($C = 0.01$ in all 10 seeds at α^* for $p = 0.95$, $n = 8000$), and the mean absolute gap to the prediction rises from 0.0018 to 0.0048 (maximum 0.059) at $n = 8000$, and to 0.010 (maximum 0.14) at $n = 500$. Using the same weights $w(\alpha)$ in the validation loss, which estimates the risk under p_α, restores agreement (maximum gap 0.013 at $n = 8000$). All E3 results above use the weighted rule; the unweighted runs (another 480 models) are kept in $\mathrm{results/e3_alpha.csv}$.

5.3 E4: an image-level check on MNIST

Construction. We keep MNIST digits 0 and 1 LeCun et al. [1998] (12 665 training images, 10% held out for validation; 2115 test images). As in Colored MNIST Arjovsky et al. [2019], the observed



(a) Training images by (label, marker) group.

(b) Accuracy against α; dashed: prediction.

Figure 5: MNIST with a corner marker. (a) The marker (blue box) agrees with the label in 95% of training images; 25% of labels are flipped. (b) Mean $\pm$ s.d. over 3 seeds.

label is the digit flipped with probability 0.25, so the digit predicts Y with accuracy $Q = 0.75$ in every environment. The shortcut is a 4×4 patch in the top-left corner, light (0.25) for $A = -1$ and dark (0.75) for $A = +1$, with $P(A = Y) = 0.95$ in training (Fig. 5a). A two-layer CNN is trained with Adam (learning rate 10^{-3}, batch 128, 10 epochs, best epoch by $w(\alpha)$-weighted validation loss) for 3 seeds.

Prediction. Since the CNN recognizes 0 versus 1 almost perfectly, the task reduces to two binary cues that agree with Y independently, the digit with probability Q and the marker with probability p. The Bayes rule follows whichever cue has the larger log-odds weight, c_Q or $|c_p|$. By Proposition 2, reweighting moves the training environment to p_α, so the model should follow the marker while $p_\alpha > 0.75$ ($\alpha < 0.44$), the digit while $0.25 < p_\alpha < 0.75$, and the inverted marker when $p_\alpha < 0.25$ ($\alpha > 1.56$). The predicted accuracies are p_{test}, 0.75 and $1 - p_{\text{test}}$ respectively (Appendix A).

Result. Figure 5b shows the three regimes. ERM agrees with the marker on 100% of test images and scores $0.950/0.492/0.100$ in the three environments (predicted $0.95/0.50/0.10$); its worst-group accuracy is 0. GB gives $0.757/0.752/0.741$ (predicted 0.75), and α^* gives 0.883 in the reversed environment (predicted 0.90). Near the two switching points ($\alpha = 0.4, 0.5, 1.5$) the change is gradual rather than a jump, with large seed-to-seed variance: the CNN is not exactly the two-cue Bayes rule, and with finite data the two cues have similar weights there.

6 Discussion and limitations

What we learned. When a shortcut is reliable in the training data, relying on it is the optimal thing to do for that data; the failure is a mismatch between environments, so more data from the same environment cannot fix it. Group reweighting works because it is a change of environment, and in our model the size of that change is linear in the reweighting strength. This gives a precise cost of robustness (for $p = 0.95$, 0.117 ID accuracy for GB) and shows that the usual choice $\alpha = 1$ is optimal only for $p_{\text{test}} = 0.5$.

Limitations. The theory assumes known group labels, a binary attribute, conditionally independent features and a known test correlation for α^*; real spurious attributes are rarely all of these. The Gaussian model is chosen so the predictions are exact, which limits external validity. E4 tests one image construction, one architecture and 3 seeds, and its prediction is qualitative near the switching points. We did not compare with Group DRO Sagawa et al. [2020a] or with methods that infer groups. Uncertainty intervals reflect seed variation in this simulation only.

7 Conclusion

In a model where the answer can be computed, the two common remedies for shortcut learning behave very differently: more data does not change OOD accuracy at all, while group reweighting moves the model along a closed-form frontier whose optimal point depends on the test environment. The same identity predicts where a CNN on MNIST switches cues.

Code. https://github.com/Er1cShi/JHU_ML_Course_Final_Project; every number and figure is produced by the scripts listed in Appendix C.

Author contributions. [Author 1] designed and ran all code and experiments and prepared the slides. [Author 2] and [Author 3]: [to be filled in, e.g. presentation, literature review, proof checking, writing]. *[Edit before submission.]*

AI-use statement. We used Claude (Anthropic), through the Claude Code agent in a project workspace, throughout the project. It decomposed the project from the course's topic description, wrote the data generator, training and analysis code, derived the propositions in Section 4, ran the experiments, produced the figures and drafted this report. Its derivations were checked by independent numerical routes in the code: Monte Carlo against (3), numerical integration against the sign approximation, and a 2-million-sample fit for the population logistic minimizer. Human judgment chose the topic, asked for the MNIST experiment and the closed-form frontier as the originality direction, and decided what to report. *[Edit before submission: describe which derivations and citations you checked by hand, what you changed or rejected, and how you prompted.]*

References

Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk minimization. *arXiv preprint arXiv:1907.02893*, 2019.

Jonathon Byrd and Zachary C. Lipton. What is the effect of importance weighting in deep learning? In *International Conference on Machine Learning (ICML)*, 2019.

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11): 665–673, 2020.

Badr Youbi Idrissi, Martin Arjovsky, Mohammad Pezeshki, and David Lopez-Paz. Simple data balancing achieves competitive worst-group-accuracy. In *Conference on Causal Learning and Reasoning (CLeaR)*, 2022.

Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.

Vaishnavh Nagarajan, Anders Andreassen, and Behnam Neyshabur. Understanding the failure modes of out-of-distribution generalization. In *International Conference on Learning Representations (ICLR)*, 2021.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

Shiori Sagawa, Pang Wei Koh, Tatsunori B. Hashimoto, and Percy Liang. Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. In *International Conference on Learning Representations (ICLR)*, 2020a.

Shiori Sagawa, Aditi Raghunathan, Pang Wei Koh, and Percy Liang. An investigation of why overparameterization exacerbates spurious correlations. In *International Conference on Machine Learning (ICML)*, 2020b.

Hidetoshi Shimodaira. Improving predictive inference under covariate shift by weighting the log-likelihood function. *Journal of Statistical Planning and Inference*, 90(2):227–244, 2000.

A Proofs

Proof of Proposition 1. Given $Y = y$, $X_s \sim N(y, 1)$ and X_z is the mixture $p\,N(y, \sigma^2) + (1 - p)\,N(-y, \sigma^2)$, independent of X_s. With $P(Y = 1) = \frac{1}{2}$,

$$\log \frac{P(Y = 1 \mid x)}{P(Y = -1 \mid x)} = \log \frac{\phi(x_s - 1)}{\phi(x_s + 1)} + \log \frac{p\,\phi_\sigma(x_z - 1) + (1 - p)\phi_\sigma(x_z + 1)}{p\,\phi_\sigma(x_z + 1) + (1 - p)\phi_\sigma(x_z - 1)} = 2x_s + g_p(x_z).$$

When x_z is far from 0 relative to σ, one mixture component dominates and $g_p(x_z) \approx c_p \operatorname{sign}(x_z)$, which gives h_p. For its accuracy in environment q, condition on $Y = 1$ (the case $Y = -1$ is symmetric). Then $\operatorname{sign}(X_z) = A$ unless $|\sigma\varepsilon_z| > 1$, which has probability at most $2\Phi(-1/\sigma)$ and changes the accuracy by at most $\Phi(-1/\sigma)$ on average over the two values of A. If $A = 1$ (probability q), h_p is correct iff $2X_s + c_p > 0$, i.e. $X_s > -c_p/2$, which has probability $\Phi(1 + c_p/2)$ since $X_s \sim N(1, 1)$. If $A = -1$ (probability $1 - q$), it is correct iff $X_s > c_p/2$, with probability $\Phi(1 - c_p/2)$. Summing gives (3). The exact Bayes rule replaces $c_p \operatorname{sign}(x_z)$ by $g_p(x_z)$; integrating over x_z numerically ($\mathrm{theory.acc_bayes_exact}$) agrees with (3) to four decimals in all our settings. □

Proof of Proposition 2. Let n_g be the group counts. Under the balance assumption, each of the two majority groups $(y, A = y)$ has $n_g = \hat{p}\,n/2$ and each minority group has $n_g = (1 - \hat{p})\,n/2$, so $n/(4n_g)$ equals $1/(2\hat{p})$ and $1/(2(1 - \hat{p}))$ respectively, and

$$w_{\text{maj}} = (1 - \alpha) + \frac{\alpha}{2\hat{p}}, \qquad w_{\text{min}} = (1 - \alpha) + \frac{\alpha}{2(1 - \hat{p})}.$$

The weighted empirical distribution puts mass $\hat{p}\,w_{\text{maj}}$ on the majority groups and $(1 - \hat{p})\,w_{\text{min}}$ on the minority groups, and does not change the distribution within a group. Since $\hat{p}\,w_{\text{maj}} + (1 - \hat{p})\,w_{\text{min}} = (1 - \alpha) + \alpha/2 + \alpha/2 = 1$, these are proper probabilities, and the majority mass is $\hat{p}(1 - \alpha) + \alpha/2 = \hat{p} - \alpha(\hat{p} - \frac{1}{2})$. Labels remain balanced by symmetry, and within-group distributions do not depend on p in (1), so the weighted distribution is that of environment p_α. Non-negativity: $w_{\text{min}} \geq 0$ for all $\alpha \geq 0$, and $w_{\text{maj}} \geq 0$ iff $\alpha \leq 2\hat{p}/(2\hat{p} - 1)$, at which point $p_\alpha = 0$. In the population ($\hat{p} = p$) the weighted risk $\mathbb{E}[w(\alpha)\,\ell]$ equals the unweighted risk in environment p_α, so the two minimizers coincide. Without exact balance the same argument applies group by group with p_α defined from the weighted group masses. □

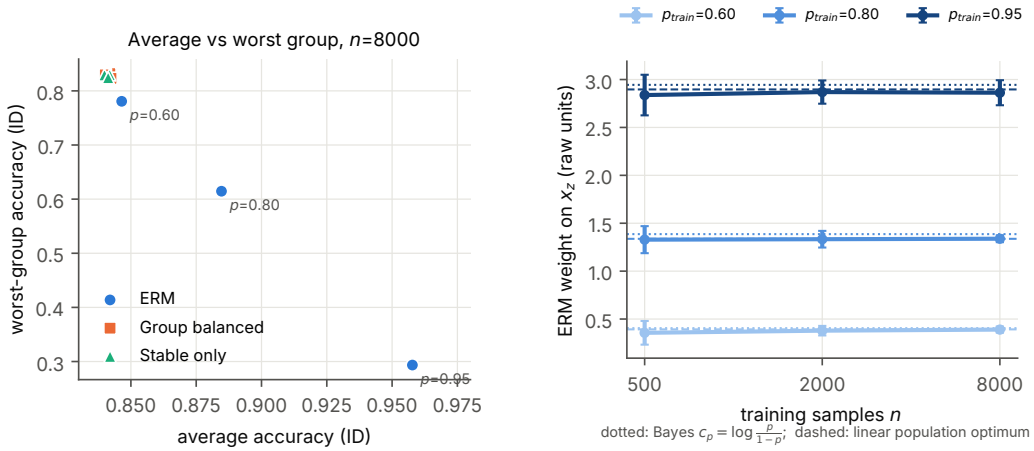
Proof of Corollary 1. Solving $p_\alpha = q$ in (4) gives $\alpha^* = (p - q)/(p - \frac{1}{2})$, which lies in the non-negative range for $q \in [0, p]$. The weighted training distribution is then the test distribution, so the weighted population risk of any predictor equals its test risk, and its minimizer over a class containing the Bayes rule is the Bayes rule of environment q, with accuracy $\mathrm{Acc}(q, q)$ by Proposition 1. □

Two-cue model for E4. Let $\tilde{d}, \tilde{a} \in \{\pm 1\}$ indicate whether the digit and the marker agree with Y; they are independent given Y, with $P(\tilde{d} = 1) = Q$ and $P(\tilde{a} = 1) = r$ in environment r. The Bayes rule predicts the label favoured by $c_Q\,\tilde{d} + c_r\,\tilde{a}$, i.e. it follows the marker if $|c_r| > c_Q$ (inverted if $r < \frac{1}{2}$) and the digit otherwise. Following the marker gives accuracy q in a test environment q, following the digit gives Q, and following the inverted marker gives $1 - q$. With $r = p_\alpha$, the switches are at $p_\alpha = Q$ and $p_\alpha = 1 - Q$, i.e. $\alpha = (p - Q)/(p - \frac{1}{2}) = 0.44$ and $\alpha = (p - 1 + Q)/(p - \frac{1}{2}) = 1.56$ for $p = 0.95$, $Q = 0.75$. □

B Population optimum of the linear model

Logistic regression is linear in x while the Bayes log-odds is a step function of x_z, so the population logistic minimizer $\beta(p) = (b_0, b_s, b_z)$ differs slightly from $(0, 2, c_p)$. We compute it by an essentially unregularized fit on 2×10^6 samples: $b_z = 0.39, 1.34, 2.90$ for $p = 0.6, 0.8, 0.95$, against $c_p = 0.41, 1.39, 2.94$, with $b_s \approx 2.0$. For any linear rule, the score given $(Y, A) = (y, a)$ is Gaussian with mean $b_0 + y\,b_s + a\,b_z$ and variance $b_s^2 + \sigma^2 b_z^2$, so its accuracy in environment q is an exact average of four normal CDF values ($\mathrm{theory.acc_linear}$). These are the predictions in Table 1 and the dashed lines in Fig. 2. The OOD accuracy of $\beta(p)$ depends only on p and q, and by standard consistency of M-estimators the fitted coefficients converge to $\beta(p)$ as $n \to \infty$.

C Additional results and implementation



(a) Average versus worst-group accuracy.

(b) Learned shortcut weight.

Figure 6: (a) ID test set, $n = 8000$. (b) ERM weight on x_z in raw units, mean $\pm$ s.d. over 10 seeds; it is close to the population value at every n.

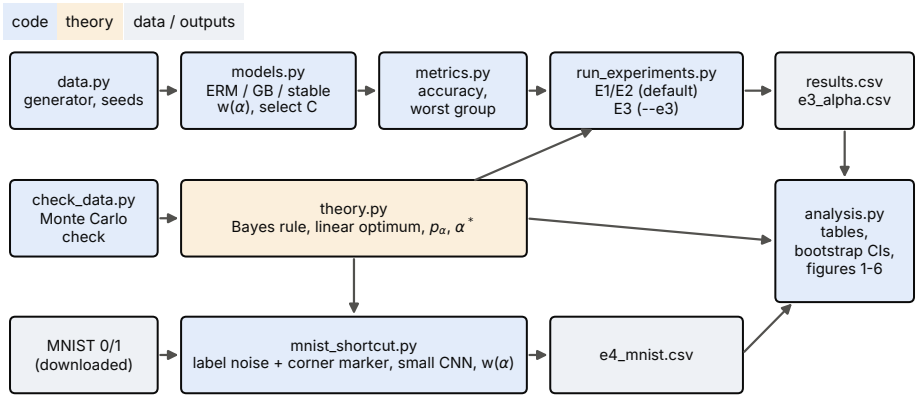


Figure 7: Code structure. Each experiment writes a CSV; analysis.py produces all tables and figures.

Implementation. Logistic regression uses scikit-learn's L-BFGS solver (tolerance 10^{-6}, at most 1000 iterations); all 1350 candidate fits in E1 and all E3 fits converged. Standardization is fitted on training data only. Seeds 0–9 index the training pool, validation and test sets; smaller training sets are prefixes of the pool. Bootstrap intervals resample the 10 seed-level paired differences 2000 times. E1 takes about 20 s and E3 about 1 min on a laptop CPU; E4 takes about 4 min.